$$C^{(12)}_{d\phi_2}\text{i1\_i2\_}\to$$

$$\frac{1}{16\pi^2}\Big(\frac{1}{4}m_1\tilde\mu g_1^4 y_d^{i1i2}LF_{2,2,0}[m_1,\tilde\mu]+\frac{1}{12}m_2\tilde\mu g_2^4(9y_d^{i1i2}-4y_d^{pi2}\delta_{pi1})LF_{2,2,0}[m_2,\tilde\mu]+$$
$$\frac{1}{6}m_2\tilde\mu g_2^4y_d^{pi2}LF_{3,2,-1}[m_2,\tilde\mu]\delta_{pi1}+\frac{1}{18}\tilde\mu g_1^2\overline{y_d}^{pr}a_d^{pr}(3y_d^{i1i2}-y_d^{si2}\delta_{si1}-2y_d^{i1s}\delta_{si2})$$
$$LF_{2,2,0}[m_{\tilde d}^r,m_{\tilde q}^p]+\frac{1}{36}\tilde\mu g_1^2\overline{y_d}^{pr}a_d^{pr}(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2})LF_{3,2,-1}[m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{1}{18}\tilde\mu g_1^2\overline{y_e}^{pr}a_e^{pr}(3y_d^{i1i2}-y_d^{si2}\delta_{si1}-2y_d^{i1s}\delta_{si2})LF_{2,2,0}[m_{\tilde e}^r,m_{\tilde l}^p]+$$
$$\frac{1}{36}\tilde\mu g_1^2\overline{y_e}^{pr}a_e^{pr}(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2})LF_{3,2,-1}[m_{\tilde e}^r,m_{\tilde l}^p]+$$
$$\frac{1}{18}\tilde\mu g_1^2\overline{y_e}^{pr}a_e^{pr}(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2})LF_{3,1,0}[m_{\tilde l}^p,m_{\tilde e}^r]+$$
$$\frac{1}{18}\tilde\mu g_1^2\overline{y_e}^{pr}a_e^{pr}(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2})LF_{3,2,-1}[m_{\tilde l}^p,m_{\tilde e}^r]-\frac{1}{8}\tilde\mu\overline{y_e}^{pr}a_e^{pr}$$
$$(g_2^2(y_d^{i1i2}+y_d^{si2}\delta_{si1})+g_1^2(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2}))LF_{4,1,-1}[m_{\tilde l}^p,m_{\tilde e}^r]+$$
$$\frac{1}{18}\tilde\mu\overline{y_e}^{pr}a_e^{pr}(3g_2^2y_d^{si2}\delta_{si1}+g_1^2(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2}))LF_{5,1,-2}[m_{\tilde l}^p,m_{\tilde e}^r]+$$
$$\frac{1}{18}\tilde\mu g_1^2\overline{y_d}^{pr}a_d^{pr}(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2})LF_{3,1,0}[m_{\tilde q}^p,m_{\tilde d}^r]+$$
$$\frac{1}{18}\tilde\mu g_1^2\overline{y_d}^{pr}a_d^{pr}(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2})LF_{3,2,-1}[m_{\tilde q}^p,m_{\tilde d}^r]-\frac{1}{24}\tilde\mu\overline{y_d}^{pr}a_d^{pr}$$
$$(9g_2^2(y_d^{i1i2}+y_d^{si2}\delta_{si1})+5g_1^2(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2}))LF_{4,1,-1}[m_{\tilde q}^p,m_{\tilde d}^r]+$$
$$\frac{1}{6}\tilde\mu\overline{y_d}^{pr}a_d^{pr}(3g_2^2y_d^{si2}\delta_{si1}+g_1^2(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2}))LF_{5,1,-2}[m_{\tilde q}^p,m_{\tilde d}^r]+$$
$$\frac{1}{36}\tilde\mu\overline{y_u}^{pr}a_u^{pr}(-9g_2^2y_d^{si2}\delta_{si1}+g_1^2(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2}))LF_{2,2,0}[m_{\tilde q}^p,m_{\tilde u}^r]+$$
$$\frac{3}{4}\tilde\mu g_2^2y_d^{i1i2}\overline{y_u}^{pr}a_u^{pr}LF_{3,1,0}[m_{\tilde q}^p,m_{\tilde u}^r]+$$
$$\frac{1}{72}\tilde\mu\overline{y_u}^{pr}a_u^{pr}((g_1^2-9g_2^2)(3y_d^{i1i2}-y_d^{si2}\delta_{si1})-2g_1^2y_d^{i1s}\delta_{si2})LF_{3,2,-1}[m_{\tilde q}^p,m_{\tilde u}^r]-$$
$$\frac{3}{4}\tilde\mu g_2^2y_d^{i1i2}\overline{y_u}^{pr}a_u^{pr}LF_{4,1,-1}[m_{\tilde q}^p,m_{\tilde u}^r]+$$
$$\frac{1}{36}\tilde\mu\overline{y_u}^{pr}a_u^{pr}(9g_2^2y_d^{si2}\delta_{si1}+g_1^2(3y_d^{i1i2}-y_d^{si2}\delta_{si1}-2y_d^{i1s}\delta_{si2}))LF_{3,1,0}[m_{\tilde u}^r,m_{\tilde q}^p]+$$
$$\frac{1}{36}\tilde\mu\overline{y_u}^{pr}a_u^{pr}(9g_2^2y_d^{si2}\delta_{si1}+g_1^2(3y_d^{i1i2}-y_d^{si2}\delta_{si1}-2y_d^{i1s}\delta_{si2}))LF_{3,2,-1}[m_{\tilde u}^r,m_{\tilde q}^p]-$$
$$\frac{1}{12}\tilde\mu\overline{y_u}^{pr}a_u^{pr}(9g_2^2y_d^{si2}\delta_{si1}+g_1^2(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2}))LF_{4,1,-1}[m_{\tilde u}^r,m_{\tilde q}^p]+$$
$$\frac{1}{6}\tilde\mu\overline{y_u}^{pr}a_u^{pr}(3g_2^2y_d^{si2}\delta_{si1}+g_1^2(-3y_d^{i1i2}+y_d^{si2}\delta_{si1}+2y_d^{i1s}\delta_{si2}))LF_{5,1,-2}[m_{\tilde u}^r,m_{\tilde q}^p]+$$
$$\frac{5}{8}m_1\tilde\mu g_1^4y_d^{i1i2}LF_{3,2,-1}[\tilde\mu,m_1]-$$
$$\frac{1}{24}m_1\tilde\mu g_1^2(3g_2^2y_d^{pi2}\delta_{pi1}+g_1^2(-3y_d^{i1i2}+y_d^{pi2}\delta_{pi1}+2y_d^{i1p}\delta_{pi2}))LF_{4,1,-1}[\tilde\mu,m_1]-$$
$$\frac{1}{2}m_1\tilde\mu g_1^4y_d^{i1i2}LF_{4,2,-2}[\tilde\mu,m_1]+$$
$$\frac{1}{18}m_1\tilde\mu g_1^2(3g_2^2y_d^{pi2}\delta_{pi1}+g_1^2(-3y_d^{i1i2}+y_d^{pi2}\delta_{pi1}+2y_d^{i1p}\delta_{pi2}))LF_{5,1,-2}[\tilde\mu,m_1]+$$
$$\frac{1}{3}m_2\tilde\mu g_2^4y_d^{pi2}LF_{3,1,0}[\tilde\mu,m_2]\delta_{pi1}+\frac{1}{24}m_2\tilde\mu g_2^4(51y_d^{i1i2}+8y_d^{pi2}\delta_{pi1})LF_{3,2,-1}[\tilde\mu,m_2]-$$
$$\frac{1}{8}m_2\tilde\mu g_2^2(7g_2^2y_d^{pi2}\delta_{pi1}+g_1^2(-3y_d^{i1i2}+y_d^{pi2}\delta_{pi1}+2y_d^{i1p}\delta_{pi2}))LF_{4,1,-1}[\tilde\mu,m_2]-$$
$$2m_2\tilde\mu g_2^4y_d^{i1i2}LF_{4,2,-2}[\tilde\mu,m_2]+$$
$$\frac{1}{6}m_2\tilde\mu g_2^2(3g_2^2y_d^{pi2}\delta_{pi1}+g_1^2(-3y_d^{i1i2}+y_d^{pi2}\delta_{pi1}+2y_d^{i1p}\delta_{pi2}))LF_{5,1,-2}[\tilde\mu,m_2]+$$
$$\frac{1}{18}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,0}[m_1,m_{\tilde d}^r,m_{\tilde q}^p]-$$
$$\frac{1}{18}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[m_1,m_{\tilde d}^r,m_{\tilde q}^p]+\frac{1}{6}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}$$
$$LF_{2,1,1,0}[m_1,m_{\tilde d}^r,\tilde\mu]-\frac{1}{6}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[m_1,m_{\tilde d}^r,\tilde\mu]-$$
$$\frac{1}{9}m_1\tilde\mu g_1^4y_d^{i1i2}LF_{3,1,1,-1}[m_1,m_{\tilde d}^{i2},\tilde\mu]+\frac{1}{12}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,0}[m_1,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{12}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[m_1,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{36}m_1\tilde\mu g_1^4y_d^{i1i2}LF_{3,1,1,-1}[m_1,m_{\tilde q}^{i1},\tilde\mu]-\frac{7}{24}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,2,1,-1}[m_1,\tilde\mu,m_{\tilde d}^r]-$$
$$\frac{7}{18}m_1\tilde\mu g_1^4y_d^{i1i2}LF_{2,2,1,-1}[m_1,\tilde\mu,m_{\tilde d}^{i2}]-\frac{5}{24}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,2,1,-1}[m_1,\tilde\mu,m_{\tilde q}^p]-$$
$$\frac{13}{72}m_1\tilde\mu g_1^4y_d^{i1i2}LF_{2,2,1,-1}[m_1,\tilde\mu,m_{\tilde q}^{i1}]+\frac{3}{4}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,0}[m_2,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{3}{4}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[m_2,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{3}{4}m_2\tilde\mu g_2^4y_d^{i1i2}LF_{3,1,1,-1}[m_2,m_{\tilde q}^{i1},\tilde\mu]+\frac{3}{2}\tilde\mu g_1^2g_2^2y_d^{i1i2}(m_1+m_2)LF_{2,2,1,-1}[m_2,\tilde\mu,m_1]-$$
$$\tilde\mu g_1^2g_2^2y_d^{i1i2}(m_1+m_2)LF_{3,2,1,-2}[m_2,\tilde\mu,m_1]-\frac{1}{8}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}$$
$$LF_{2,2,1,-1}[m_2,\tilde\mu,m_{\tilde d}^r]-\frac{9}{8}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,2,1,-1}[m_2,\tilde\mu,m_{\tilde q}^p]+$$
$$\frac{1}{8}m_2\tilde\mu g_2^4(-17y_d^{i1i2}+2y_d^{pi2}\delta_{pi1})LF_{2,2,1,-1}[m_2,\tilde\mu,m_{\tilde q}^{i1}]-$$
$$\frac{1}{4}m_2\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{2,2,1,-1}[m_2,\tilde\mu,m_{\tilde u}^r]-\frac{4}{3}m_3\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}$$
$$LF_{2,1,1,0}[m_3,m_{\tilde d}^r,m_{\tilde q}^p]+\frac{4}{3}m_3\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[m_3,m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{1}{4}\tilde\mu y_d^{i1i2}\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{sr}a_e^{pt}LF_{2,2,1,-1}[m_{\tilde l}^p,m_{\tilde l}^s,m_{\tilde e}^t]+\frac{3}{4}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{sr}y_d^{i1i2}a_d^{pt}$$
$$LF_{2,2,1,-1}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde d}^t]-\frac{3}{4}\tilde\mu\overline{y_d}^{pr}y_d^{sr}y_d^{i1i2}\overline{y_u}^{st}a_u^{pt}LF_{2,1,1,0}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t]+$$
$$\frac{3}{4}\tilde\mu\overline{y_d}^{pr}y_d^{sr}y_d^{i1i2}\overline{y_u}^{st}a_u^{pt}LF_{3,1,1,-1}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t]+\frac{3}{8}\tilde\mu\overline{y_d}^{pr}y_d^{sr}y_d^{i1i2}\overline{y_u}^{st}a_u^{pt}$$
$$LF_{2,2,1,-1}[m_{\tilde q}^p,m_{\tilde u}^t,m_{\tilde q}^s]-\frac{3}{4}\tilde\mu\overline{y_d}^{pr}y_d^{sr}y_d^{i1i2}\overline{y_u}^{st}a_u^{pt}LF_{2,1,1,0}[m_{\tilde q}^s,m_{\tilde q}^p,m_{\tilde u}^t]+$$
$$\frac{3}{4}\tilde\mu\overline{y_d}^{pr}y_d^{sr}y_d^{i1i2}\overline{y_u}^{st}a_u^{pt}LF_{3,1,1,-1}[m_{\tilde q}^s,m_{\tilde q}^p,m_{\tilde u}^t]+\frac{3}{8}\tilde\mu\overline{y_d}^{pr}y_d^{sr}y_d^{i1i2}\overline{y_u}^{st}$$
$$a_u^{pt}LF_{2,2,1,-1}[m_{\tilde q}^s,m_{\tilde u}^t,m_{\tilde q}^p]+\frac{1}{4}\tilde\mu g_1^2g_2^2y_d^{i1i2}(m_1+7m_2)LF_{3,1,1,-1}[\tilde\mu,m_1,m_2]-$$
$$\frac{3}{2}m_2\tilde\mu g_1^2g_2^2y_d^{i1i2}LF_{4,1,1,-2}[\tilde\mu,m_1,m_2]+\frac{1}{6}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,0}[\tilde\mu,m_1,m_{\tilde d}^r]-$$
$$\frac{5}{12}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[\tilde\mu,m_1,m_{\tilde d}^r]+$$
$$\frac{1}{12}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,0}[\tilde\mu,m_1,m_{\tilde q}^p]-\frac{1}{3}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}$$
$$LF_{3,1,1,-1}[\tilde\mu,m_1,m_{\tilde q}^p]-\frac{1}{2}\tilde\mu g_1^2g_2^2y_d^{i1i2}(m_1+3m_2)LF_{3,2,1,-2}[\tilde\mu,m_2,m_1]-$$
$$\frac{1}{4}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[\tilde\mu,m_2,m_{\tilde d}^r]+\frac{3}{4}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}$$
$$LF_{2,1,1,0}[\tilde\mu,m_2,m_{\tilde q}^p]-\frac{3}{2}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{3,1,1,-1}[\tilde\mu,m_2,m_{\tilde q}^p]-$$
$$\frac{1}{2}m_2\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{3,1,1,-1}[\tilde\mu,m_2,m_{\tilde u}^r]-\frac{1}{2}\tilde\mu\overline{y_d}^{pr}y_d^{si2}y_d^{i1r}\overline{y_u}^{st}a_u^{pt}$$
$$LF_{2,1,1,0}[\tilde\mu,m_{\tilde q}^p,m_{\tilde u}^t]+\frac{1}{2}\tilde\mu\overline{y_d}^{pr}y_d^{si2}y_d^{i1r}\overline{y_u}^{st}a_u^{pt}LF_{3,1,1,-1}[\tilde\mu,m_{\tilde q}^p,m_{\tilde u}^t]-$$
$$\frac{1}{12}m_1\tilde\mu g_1^2g_2^2y_d^{i1i2}LF_{2,1,1,1,-1}[m_1,m_2,m_{\tilde q}^{i1},\tilde\mu]+\frac{1}{8}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}$$
$$LF_{1,1,1,1,0}[m_1,m_{\tilde d}^r,m_{\tilde q}^p,\tilde\mu]-\frac{3}{16}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[m_1,m_{\tilde d}^r,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{36}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{1,1,1,1,0}[m_1,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]+\frac{1}{72}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}$$
$$LF_{2,1,1,1,-1}[m_1,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]-\frac{1}{24}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{1,1,1,1,0}[m_1,m_{\tilde d}^r,m_{\tilde q}^{i1},\tilde\mu]+$$
$$\frac{1}{48}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_1,m_{\tilde d}^r,m_{\tilde q}^{i1},\tilde\mu]-\frac{1}{9}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}$$
$$LF_{1,1,1,1,0}[m_1,m_{\tilde d}^{i2},m_{\tilde d}^r,m_{\tilde q}^p]+\frac{1}{18}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[m_1,m_{\tilde d}^{i2},m_{\tilde d}^r,m_{\tilde q}^p]-$$
$$\frac{1}{12}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{1,1,1,1,0}[m_1,m_{\tilde d}^{i2},m_{\tilde q}^p,\tilde\mu]+\frac{1}{24}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}$$
$$LF_{2,1,1,1,-1}[m_1,m_{\tilde d}^{i2},m_{\tilde q}^p,\tilde\mu]+\frac{1}{9}\tilde\mu g_1^4a_d^{i1i2}LF_{2,1,1,1,-1}[m_1,m_{\tilde d}^{i2},m_{\tilde q}^{i1},\tilde\mu]-$$
$$\frac{1}{12}m_2\tilde\mu g_1^2g_2^2y_d^{i1i2}LF_{2,1,1,1,-1}[m_2,m_1,m_{\tilde q}^{i1},\tilde\mu]+\frac{1}{8}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}$$
$$LF_{1,1,1,1,0}[m_2,m_{\tilde d}^r,m_{\tilde q}^p,\tilde\mu]-\frac{3}{16}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[m_2,m_{\tilde d}^r,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{4}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{1,1,1,1,0}[m_2,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]+\frac{1}{8}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}$$
$$LF_{2,1,1,1,-1}[m_2,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]-\frac{1}{8}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{1,1,1,1,0}[m_2,m_{\tilde d}^r,m_{\tilde q}^{i1},\tilde\mu]+$$
$$\frac{1}{16}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_2,m_{\tilde d}^r,m_{\tilde q}^{i1},\tilde\mu]+\frac{1}{4}m_2\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}$$
$$LF_{1,1,1,1,0}[m_2,m_{\tilde q}^p,m_{\tilde u}^r,\tilde\mu]-\frac{3}{8}m_2\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{2,1,1,1,-1}[m_2,m_{\tilde q}^p,m_{\tilde u}^r,\tilde\mu]-$$
$$\frac{1}{2}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{1,1,1,1,0}[m_2,m_{\tilde q}^{i1},m_{\tilde q}^p,m_{\tilde u}^r]+\frac{1}{4}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}$$
$$LF_{2,1,1,1,-1}[m_2,m_{\tilde q}^{i1},m_{\tilde q}^p,m_{\tilde u}^r]-\frac{1}{4}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{1,1,1,1,0}[m_2,m_{\tilde q}^{i1},m_{\tilde u}^r,\tilde\mu]+$$
$$\frac{1}{8}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{2,1,1,1,-1}[m_2,m_{\tilde q}^{i1},m_{\tilde u}^r,\tilde\mu]-\frac{4}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}$$
$$LF_{1,1,1,1,0}[m_3,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]+\frac{2}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_3,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]-$$
$$\frac{4}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{1,1,1,1,0}[m_3,m_{\tilde d}^{i2},m_{\tilde d}^r,m_{\tilde q}^p]+\frac{2}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}$$
$$LF_{2,1,1,1,-1}[m_3,m_{\tilde d}^{i2},m_{\tilde d}^r,m_{\tilde q}^p]-\frac{1}{18}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[m_{\tilde d}^r,m_1,m_{\tilde d}^{i2},m_{\tilde q}^p]+$$
$$\frac{1}{48}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde d}^r,m_1,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{48}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde d}^r,m_1,m_{\tilde q}^{i1},\tilde\mu]-$$
$$\frac{1}{16}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde d}^r,m_2,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{16}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde d}^r,m_2,m_{\tilde q}^{i1},\tilde\mu]-$$
$$\frac{2}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[m_{\tilde d}^r,m_3,m_{\tilde d}^{i2},m_{\tilde q}^p]-$$
$$\frac{1}{2}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pi2}y_d^{i1t}a_d^{sr}LF_{1,1,1,1,0}[m_{\tilde d}^r,m_{\tilde d}^t,m_{\tilde q}^s,\tilde\mu]-$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pi2}y_d^{i1t}a_d^{sr}LF_{2,1,1,1,-1}[m_{\tilde d}^r,m_{\tilde d}^t,m_{\tilde q}^s,\tilde\mu]+$$
$$\frac{3}{4}\tilde\mu\overline{y_d}^{st}y_d^{i1i2}\overline{a_d}^{pr}a_d^{pt}a_d^{sr}LF_{2,2,1,1,-1}[m_{\tilde d}^r,m_{\tilde q}^p,m_{\tilde d}^t,m_{\tilde q}^s]+$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pi2}y_d^{i1t}a_d^{sr}LF_{2,1,1,1,-1}[m_{\tilde d}^t,m_{\tilde d}^r,m_{\tilde q}^s,\tilde\mu]-$$
$$\frac{3}{4}\tilde\mu\overline{y_d}^{st}y_d^{i1i2}\overline{a_d}^{pr}a_d^{pt}a_d^{sr}LF_{2,2,1,1,-1}[m_{\tilde d}^t,m_{\tilde q}^p,m_{\tilde d}^r,m_{\tilde q}^s]-$$
$$\frac{1}{2}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{si2}y_d^{i1r}a_d^{pt}LF_{1,1,1,1,0}[m_{\tilde d}^t,m_{\tilde q}^p,m_{\tilde q}^s,\tilde\mu]+$$
$$\frac{1}{18}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[m_{\tilde d}^{i2},m_1,m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{1}{24}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[m_{\tilde d}^{i2},m_1,m_{\tilde q}^p,\tilde\mu]+$$
$$\frac{2}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[m_{\tilde d}^{i2},m_3,m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{1}{4}\tilde\mu y_d^{i1i2}\overline{y_e}^{st}\overline{a_e}^{pr}a_e^{pt}a_e^{sr}LF_{2,2,1,1,-1}[m_{\tilde e}^r,m_{\tilde l}^p,m_{\tilde e}^t,m_{\tilde l}^s]-$$
$$\frac{1}{4}\tilde\mu y_d^{i1i2}\overline{y_e}^{st}\overline{a_e}^{pr}a_e^{pt}a_e^{sr}LF_{2,2,1,1,-1}[m_{\tilde e}^t,m_{\tilde l}^p,m_{\tilde e}^r,m_{\tilde l}^s]+$$
$$\frac{1}{2}\tilde\mu y_d^{i1i2}\overline{y_e}^{st}\overline{a_e}^{pr}a_e^{pt}a_e^{sr}LF_{2,1,1,1,0}[m_{\tilde l}^p,m_{\tilde e}^r,m_{\tilde e}^t,m_{\tilde l}^s]-$$
$$\frac{1}{2}\tilde\mu y_d^{i1i2}\overline{y_e}^{st}\overline{a_e}^{pr}a_e^{pt}a_e^{sr}LF_{3,1,1,1,-1}[m_{\tilde l}^p,m_{\tilde e}^r,m_{\tilde e}^t,m_{\tilde l}^s]-$$
$$\frac{1}{72}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_1,m_{\tilde d}^r,m_{\tilde q}^{i1}]-$$
$$\frac{1}{48}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_1,m_{\tilde d}^r,\tilde\mu]-$$
$$\frac{1}{24}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_1,m_{\tilde d}^{i2},\tilde\mu]-$$
$$\frac{1}{8}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_2,m_{\tilde d}^r,m_{\tilde q}^{i1}]+$$
$$\frac{1}{16}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_2,m_{\tilde d}^r,\tilde\mu]-$$
$$\frac{1}{4}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_2,m_{\tilde q}^{i1},m_{\tilde u}^r]+$$
$$\frac{1}{8}m_2\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_2,m_{\tilde u}^r,\tilde\mu]-$$
$$\frac{2}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_3,m_{\tilde d}^r,m_{\tilde q}^{i1}]+$$
$$\frac{3}{2}\tilde\mu\overline{y_d}^{st}y_d^{i1i2}\overline{a_d}^{pr}a_d^{pt}a_d^{sr}LF_{2,1,1,1,0}[m_{\tilde q}^p,m_{\tilde d}^r,m_{\tilde d}^t,m_{\tilde q}^s]-$$
$$\frac{3}{2}\tilde\mu\overline{y_d}^{st}y_d^{i1i2}\overline{a_d}^{pr}a_d^{pt}a_d^{sr}LF_{3,1,1,1,-1}[m_{\tilde q}^p,m_{\tilde d}^r,m_{\tilde d}^t,m_{\tilde q}^s]-$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{si2}y_d^{i1r}a_d^{pt}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_{\tilde d}^t,m_{\tilde q}^s,\tilde\mu]+$$
$$\frac{3}{4}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,1,1,1,0}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t,m_{\tilde u}^r]-$$
$$\frac{3}{4}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{3,1,1,1,-1}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t,m_{\tilde u}^r]-$$
$$\frac{1}{2}\tilde\mu\overline{y_d}^{pr}y_d^{si2}y_d^{i1r}\overline{y_u}^{st}a_u^{pt}LF_{1,1,1,1,0}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t,\tilde\mu]-$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}y_d^{si2}y_d^{i1r}\overline{y_u}^{st}a_u^{pt}LF_{2,1,1,1,-1}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t,\tilde\mu]-$$
$$\frac{9}{8}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,2,1,1,-1}[m_{\tilde q}^p,m_{\tilde u}^r,m_{\tilde q}^s,m_{\tilde u}^t]-$$
$$\frac{3}{8}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,2,1,1,-1}[m_{\tilde q}^p,m_{\tilde u}^t,m_{\tilde q}^s,m_{\tilde u}^r]+$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{si2}y_d^{i1r}a_d^{pt}LF_{2,1,1,1,-1}[m_{\tilde q}^s,m_{\tilde d}^t,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{3}{4}\tilde\mu y_d^{i1i2}\overline{y_u}^{st}a_d^{sr}(\overline{a_d}^{pr}a_u^{pt}+\tilde\mu^2\overline{y_d}^{pr}y_u^{pt})LF_{2,2,1,1,-1}[m_{\tilde q}^s,m_{\tilde q}^p,m_{\tilde d}^r,m_{\tilde u}^t]+$$
$$\frac{3}{4}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,1,1,1,0}[m_{\tilde q}^s,m_{\tilde q}^p,m_{\tilde u}^t,m_{\tilde u}^r]-$$
$$\frac{3}{4}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{3,1,1,1,-1}[m_{\tilde q}^s,m_{\tilde q}^p,m_{\tilde u}^t,m_{\tilde u}^r]+$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}y_d^{si2}y_d^{i1r}\overline{y_u}^{st}a_u^{pt}LF_{2,1,1,1,-1}[m_{\tilde q}^s,m_{\tilde q}^p,m_{\tilde u}^t,\tilde\mu]+$$
$$\frac{3}{8}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,2,1,1,-1}[m_{\tilde q}^s,m_{\tilde u}^r,m_{\tilde q}^p,m_{\tilde u}^t]-$$
$$\frac{1}{2}\tilde\mu y_d^{pi2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{i1t}a_u^{sr}LF_{1,1,1,1,0}[m_{\tilde q}^s,m_{\tilde u}^r,m_{\tilde u}^t,\tilde\mu]-$$
$$\frac{3}{8}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,2,1,1,-1}[m_{\tilde q}^s,m_{\tilde u}^t,m_{\tilde q}^p,m_{\tilde u}^r]+$$
$$\frac{1}{72}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^{i1},m_1,m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{1}{48}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^{i1},m_1,m_{\tilde d}^r,\tilde\mu]+$$
$$\frac{1}{8}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^{i1},m_2,m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{1}{16}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^{i1},m_2,m_{\tilde d}^r,\tilde\mu]+$$
$$\frac{1}{4}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^{i1},m_2,m_{\tilde q}^p,m_{\tilde u}^r]+$$
$$\frac{1}{8}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^{i1},m_2,m_{\tilde u}^r,\tilde\mu]+$$
$$\frac{2}{3}\tilde\mu g_3^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[m_{\tilde q}^{i1},m_3,m_{\tilde d}^r,m_{\tilde q}^p]-$$
$$\frac{1}{8}m_2\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{2,1,1,1,-1}[m_{\tilde u}^r,m_2,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{8}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{2,1,1,1,-1}[m_{\tilde u}^r,m_2,m_{\tilde q}^{i1},\tilde\mu]+$$
$$\frac{3}{2}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,1,1,1,0}[m_{\tilde u}^r,m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t]-$$
$$\frac{3}{2}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{3,1,1,1,-1}[m_{\tilde u}^r,m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t]-$$
$$\frac{1}{4}\tilde\mu y_d^{pi2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{i1t}a_u^{sr}LF_{2,1,1,1,-1}[m_{\tilde u}^r,m_{\tilde q}^s,m_{\tilde u}^t,\tilde\mu]+$$
$$\frac{1}{4}\tilde\mu y_d^{pi2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{i1t}a_u^{sr}LF_{2,1,1,1,-1}[m_{\tilde u}^t,m_{\tilde q}^s,m_{\tilde u}^r,\tilde\mu]-$$
$$\frac{3}{4}\tilde\mu^3y_d^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,2,1,1,-1}[m_{\tilde u}^t,m_{\tilde u}^r,m_{\tilde q}^p,m_{\tilde q}^s]-$$
$$\frac{1}{3}m_2\tilde\mu g_1^2g_2^2y_d^{i1i2}LF_{2,1,1,1,-1}[\tilde\mu,m_1,m_2,m_{\tilde d}^{i2}]-$$
$$\frac{1}{24}\tilde\mu g_1^2g_2^2y_d^{i1i2}(13m_1+5m_2)LF_{2,1,1,1,-1}[\tilde\mu,m_1,m_2,m_{\tilde q}^{i1}]-$$
$$\frac{9}{16}m_1\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[\tilde\mu,m_1,m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{1}{16}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[\tilde\mu,m_1,m_{\tilde d}^r,m_{\tilde q}^{i1}]+$$
$$\frac{1}{8}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{2,1,1,1,-1}[\tilde\mu,m_1,m_{\tilde d}^{i2},m_{\tilde q}^p]-$$
$$\frac{9}{16}m_2\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,1,1,-1}[\tilde\mu,m_2,m_{\tilde d}^r,m_{\tilde q}^p]+$$
$$\frac{3}{16}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{2,1,1,1,-1}[\tilde\mu,m_2,m_{\tilde d}^r,m_{\tilde q}^{i1}]-$$
$$\frac{9}{8}m_2\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{2,1,1,1,-1}[\tilde\mu,m_2,m_{\tilde q}^p,m_{\tilde u}^r]+$$
$$\frac{3}{8}\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{2,1,1,1,-1}[\tilde\mu,m_2,m_{\tilde q}^{i1},m_{\tilde u}^r]+$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pi2}y_d^{i1t}a_d^{sr}LF_{2,1,1,1,-1}[\tilde\mu,m_{\tilde d}^r,m_{\tilde d}^t,m_{\tilde q}^s]+$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{si2}y_d^{i1r}a_d^{pt}LF_{2,1,1,1,-1}[\tilde\mu,m_{\tilde d}^t,m_{\tilde q}^p,m_{\tilde q}^s]+$$
$$\frac{1}{4}\tilde\mu\overline{y_d}^{pr}y_d^{si2}y_d^{i1r}\overline{y_u}^{st}a_u^{pt}LF_{2,1,1,1,-1}[\tilde\mu,m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^t]+$$
$$\frac{1}{4}\tilde\mu y_d^{pi2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{i1t}a_u^{sr}LF_{2,1,1,1,-1}[\tilde\mu,m_{\tilde q}^s,m_{\tilde u}^r,m_{\tilde u}^t]+$$
$$\frac{1}{3}\tilde\mu g_1^2g_2^2a_d^{i1i2}LF_{1,1,1,1,1,-1}[m_1,m_2,m_{\tilde d}^{i2},m_{\tilde q}^{i1},\tilde\mu]+$$
$$\frac{1}{6}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{1,1,1,1,1,-1}[m_1,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p,\tilde\mu]+$$
$$\frac{1}{3}\tilde\mu g_1^2\overline{y_d}^{pr}y_d^{i1r}a_d^{pi2}LF_{1,1,1,1,1,-1}[m_1,m_{\tilde d}^{i2},m_{\tilde d}^r,m_{\tilde q}^p,\tilde\mu]-$$
$$\frac{1}{9}m_1\tilde\mu g_1^2\overline{y_d}^{pr}a_d^{pi2}a_d^{i1r}LF_{1,1,1,1,1,0}[m_1,m_{\tilde d}^{i2},m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]+$$
$$\frac{1}{2}\tilde\mu g_2^2\overline{y_d}^{pr}y_d^{pi2}a_d^{i1r}LF_{1,1,1,1,1,-1}[m_2,m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p,\tilde\mu]+$$
$$\tilde\mu g_2^2y_d^{pi2}\overline{y_u}^{pr}a_u^{i1r}LF_{1,1,1,1,1,-1}[m_2,m_{\tilde q}^{i1},m_{\tilde q}^p,m_{\tilde u}^r,\tilde\mu]+$$
$$\frac{8}{3}m_3\tilde\mu g_3^2\overline{y_d}^{pr}a_d^{pi2}a_d^{i1r}LF_{1,1,1,1,1,0}[m_3,m_{\tilde d}^{i2},m_{\tilde d}^r,m_{\tilde q}^{i1},m_{\tilde q}^p]+$$
$$\tilde\mu^3y_d^{si2}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{i1r}a_u^{pt}LF_{1,1,1,1,1,0}[m_{\tilde q}^p,m_{\tilde q}^s,m_{\tilde u}^r,m_{\tilde u}^t,\tilde\mu]\Big)$$